



TP CompAL

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Difficulty: Medium

Description

Mr. CompAL is someone who loves to repeat the same example and exercise in every knowledge domain he works on, just with a slight twist each time. You've probably seen the same expression tree "*If you know, you know*" at least five times by now, from data structures to automata theory, and yes, even sprinkled into operating systems for no reason.

But hey, that repetition made something click! It's exactly what compilers do too, **repeating structured patterns**, parsing symbols, building trees, and evaluating them with mechanical precision. That brings us to a really funny realization!

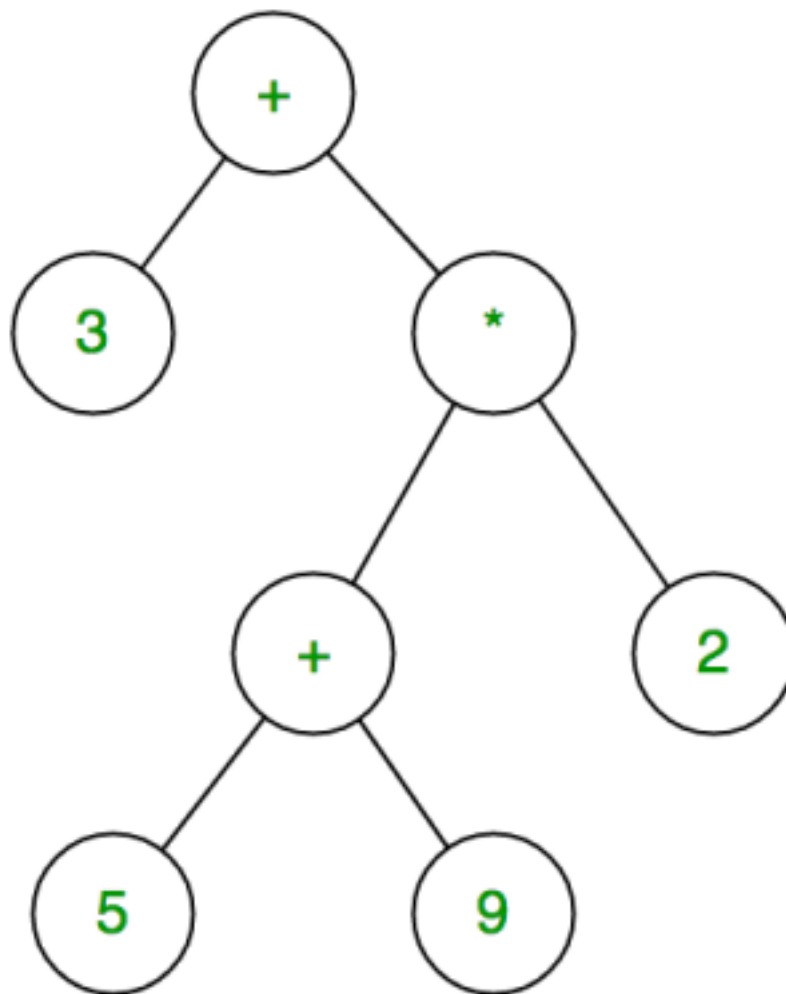
As we know "*and if you don't, here's to ya!*" **compilers**, when dealing with arithmetic and logical expressions, rely heavily on a module called the **expression evaluator**. This module traverses what's known as an **Abstract Syntax Tree (AST)**—which looks a lot like that binary tree you've seen a million times—evaluating the operations based on the grammar rules defined in the language.

And so, inspired by the never-ending reign of Mr. CompAL favorite example, here comes the challenge, just compiler-flavored

Simulate the behavior of a mini expression evaluator module from a compiler's backend by recursively computing the result of an



arithmetic expression represented as a binary tree, A tree that represents the following



This is a sample example of the express $(3 + ((5 + 9) * 2))$ which gives 31.

Inputs

String formatted evaluation tree for an arithmetic expression with parentheses around each sub evaluation.



Outputs

The final result is the evaluation output of the input tree formatted string after all the nodes evaluated.

Constraints

The output should always display floats with two digits after the decimal point.

Examples

Input	Output
<code>('*', ('+', 3, 2), ('+', 4, 5))</code>	45.00
<code>('/', ('+', ('*', 6, ('-', 9, 3)), ('/', 8, 2)), ('-', 10, 4))</code>	6.67
<code>('*', ('+', 3, ('*', 2, 5)), ('-', 20, ('/', 8, 2)))</code>	208.00